

Question 1

Friday, 15 September 2017 4:04 PM

Question 1 - (5 marks)

Figure 1 shows an unlabelled P-v diagram for a pure substance. Fill in the following information on the diagram:

a) Phase regions:

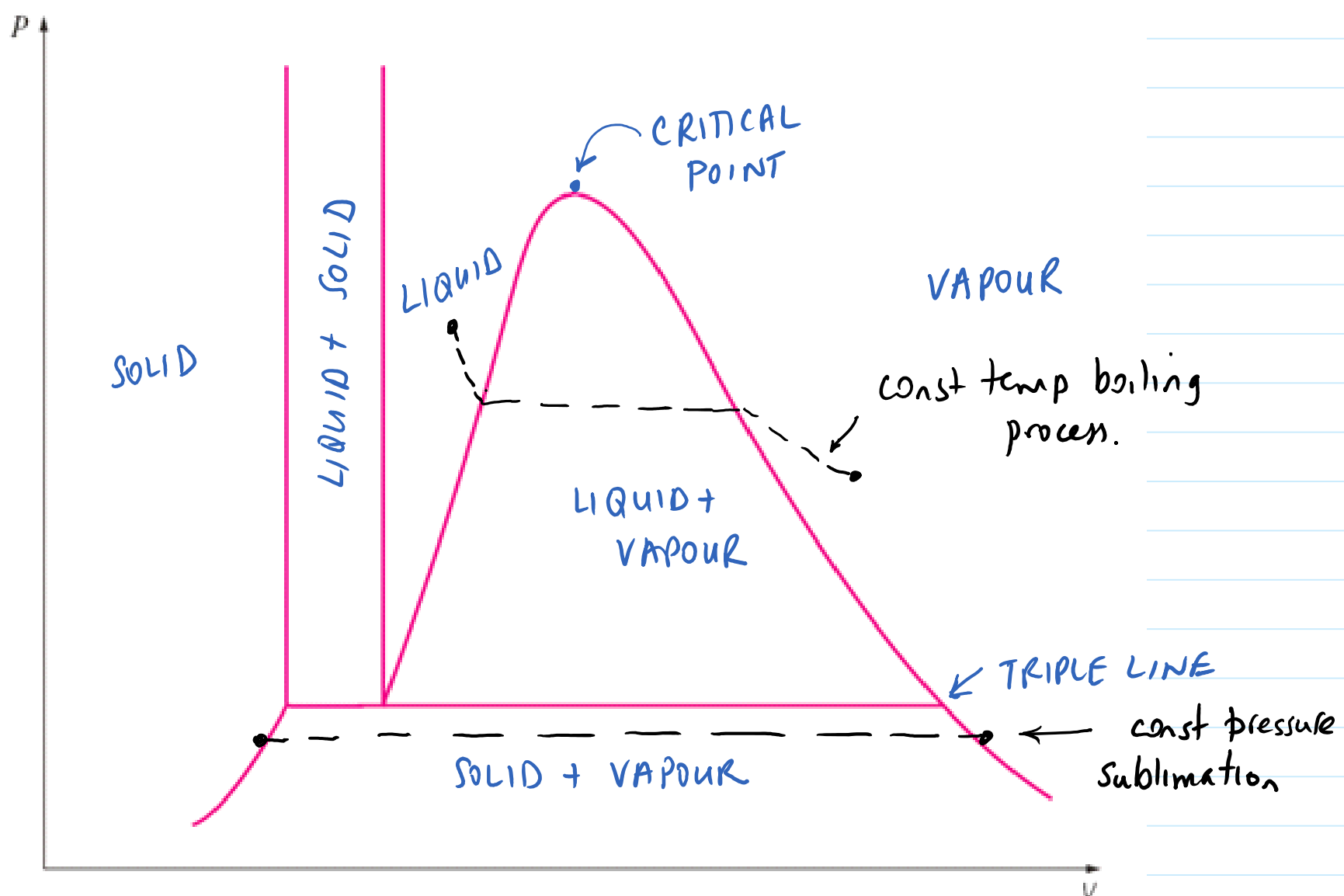
- Liquid
- Solid
- Vapour
- Liquid + Solid
- Solid + Vapour
- Liquid + Vapour.

b) Important features:

- Critical point
- Triple Line.

c) Process Lines: (clearly identify/label each process)

- A constant pressure sublimation process
- A constant temperature boiling process. (from compressed liquid to superheated vapour)



$\frac{1}{2}$ mark
each
 $\times 10$
 $= 5$

Question 2

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Question 2 - (16 marks)

A piston-cylinder device (Figure 2), whose piston is resting on a set of stops, provides an initial volume that contains 2 kg of steam at 300 kPa and a quality of $x = 0.25$. Heat is now transferred to the steam until its pressure reaches 500 kPa at which point the piston begins to move. Heat continues to be transferred to the steam (the piston maintains a constant pressure of 500 kPa) until the volume increases to twice the initial volume. Determine:

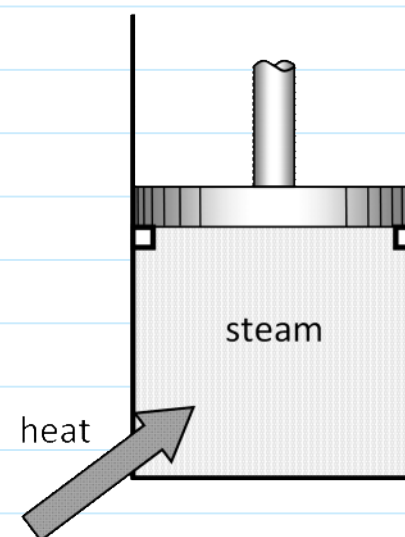


Figure 1

- The initial volume.
- The work done by the expanding steam.
- The total heat transferred to the steam during this process.

Show the process on a P - v diagram (with respect to saturation lines).

(a) Initial conditions: State 1: $m = 2 \text{ kg}$, $p = 300 \text{ kPa}$, $x = 0.25$

$$V_1 = m v_1, \quad v_1 = v_f + x v_{fg} = 0.001073 + 0.25(0.60582 - 0.001073)$$

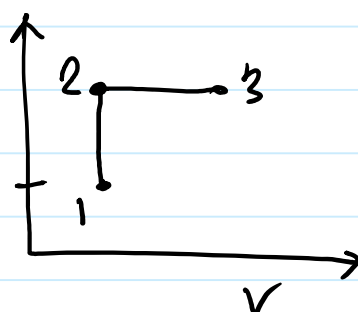
$$\Rightarrow v_1 = 0.15226 \text{ m}^3/\text{kg}$$

$$\Rightarrow V_1 = 2 \times 0.15226 = 0.30452 \text{ m}^3 \quad (\text{initial volume})$$

2

(b) $W = W_b = \int P dV \rightarrow$ area under P - V diagram

basic process



1 $\rightarrow$ 2 = const. volume

2 $\rightarrow$ 3 = const. pressure

$$\Rightarrow W_b = W_{13} = W_{12} + W_{23} = 0 + P_2 (v_3 - v_2)$$

Now, State 2: $v_2 = v_1 = 0.30452 \text{ m}^3$, $P_2 = 500 \text{ kPa}$

State 3: $v_3 = 2 \times v_2 = 2 \times 0.30452 = 0.60904 \text{ m}^3$

2

$$P_3 = P_2 = 500 \text{ kPa}$$

$$\Rightarrow W_b = W_{13} = 500(0.60904 - 0.30452)$$

$$= +152.26 \text{ kJ}$$

2

Question 2 (cont'd)

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(c) First Law - closed system (energy balance) $Q - W = \Delta U$

$$\Rightarrow Q_{13} - W_{13} = m(u_3 - u_1)$$

need u_1 and u_3

$$u_1 = u_f + x u_{fg} @ 300 \text{ kPa} = 561.11 + 0.25 \times 1982.1 = 1056.635 \text{ kJ/kg}$$

$$u_3 \dots \text{ need } x_3, \quad v_3 = v_2 = 2v_1 = 2 \times 0.15226 = 0.30452 \text{ m}^3/\text{kg}$$

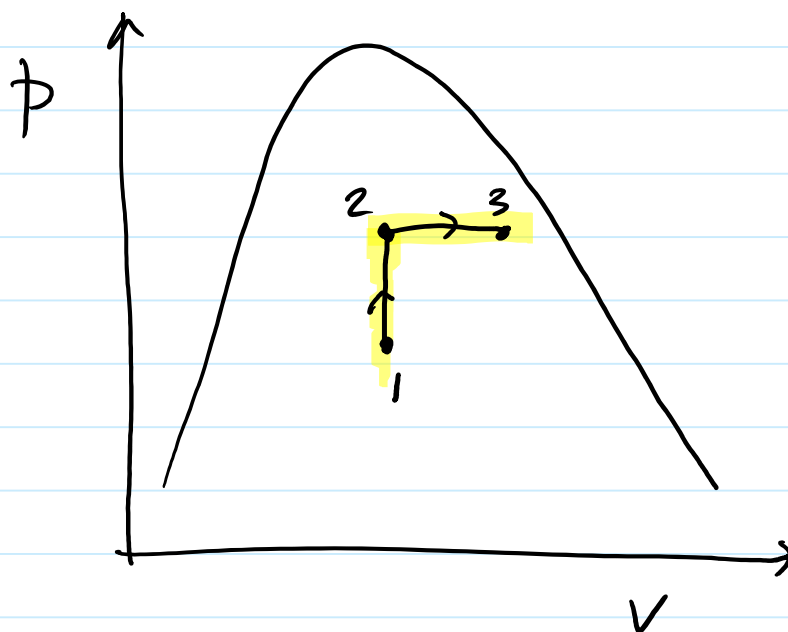
$$\text{at } 500 \text{ kPa} \quad v_f = 0.001093 \text{ m}^3/\text{kg} \text{ and } v_g = 0.37483 \text{ m}^3/\text{kg}$$

$\Rightarrow$ saturated mixture since $v_f < v_3 < v_g$

$$\text{So } x_3 = \frac{v_3 - v_f}{v_{fg}} = \frac{0.30452 - 0.001093}{0.37483 - 0.001093} = 0.812$$

$$\Rightarrow u_3 = u_f + x u_{fg} @ 500 \text{ kPa} = 639.54 + 0.812 \times 1921.2 = 2199.31 \text{ kJ/kg}$$

$$\begin{aligned} \Rightarrow Q_{13} &= W_{13} + m(u_3 - u_1) \\ &= +152.26 + 2(2199.31 - 1056.35) \\ &= 152.26 + 2285.35 \\ &= \underline{\underline{+2437.6 \text{ J}}} \end{aligned}$$



Question 3

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Question 3 - (10 marks)

A 5 m x 6 m x 3 m room at atmospheric pressure is heated by the radiator of a steam heating system (Figure 3). The steam radiator transfers heat at a rate of 15,000 kJ/hour and a 200 W fan is used to distribute the warm air around the room. The rate of heat loss from the room is estimated to be around 100 kJ/min. The initial temperature of the room is 7°C, and it is desired to heat the room to 22°C.

- Determine how long it will take to heat the air in the room assuming the room is a *constant volume* system. Also, state any other assumptions you make.
- Is it reasonable to assume that the room is a constant volume system? Justify your answer.

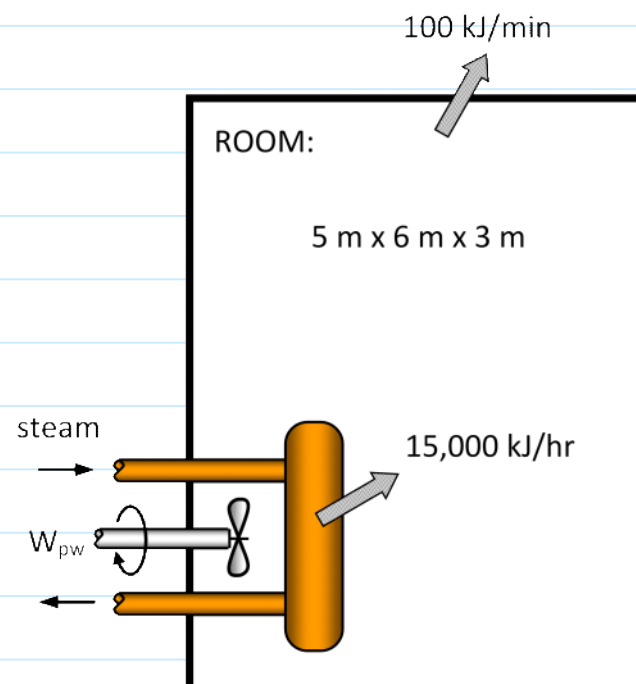


Figure 1

(a) First Law - closed system (energy balance)

$$Q - W = \Delta U = m(u_2 - u_1) = m C_v \Delta T$$

$$\Rightarrow (\dot{Q} - \dot{W}) \Delta t = m(u_2 - u_1) = m C_v \Delta T$$

Need mass: ideal gas

initial conditions:

$$p_1 = \text{atmospheric} = 101.3 \text{ kPa}$$

$$V_1 = 5 \times 6 \times 3 = 90 \text{ m}^3$$

$$T_1 = 7^\circ\text{C} = 280 \text{ K}$$

$$\Rightarrow m = \frac{p_1 V_1}{R T_1} = \frac{101.3 \times 90}{0.287 \times 280} = 113.45 \text{ kg}$$

If we use Table A-17: $u_1 @ 280 \text{ K} = 199.75 \text{ kJ/kg}$

final state: $T_2 = 22^\circ\text{C} = 295 \text{ K} \Rightarrow u_2 = 210.49 \text{ kJ/kg}$

$$\Rightarrow (\dot{Q} - \dot{W}) \Delta t = m(u_2 - u_1)$$

$$\Rightarrow \left[\frac{+15,000 \text{ kJ}}{3600 \text{ s}} - \frac{100 \text{ kJ}}{60 \text{ s}} - \left(- \frac{200 \text{ W}}{1000 \text{ W}} \cdot \frac{\text{kJ}}{\text{s}} \right) \right] \cdot \Delta t$$

$$= 113.45 (210.49 - 199.75)$$

Question 3 (cont'd)

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$$\Rightarrow 2.7 \frac{\text{kJ}}{\text{s}} \cdot \Delta t = 1218.47 \text{ kJ}$$

$$\Rightarrow \Delta t = 451.3 \text{ s}$$

$$= \underline{\underline{7.52 \text{ minutes}}}$$

2

OR if use avg C_v : $T_{\text{avg}} = \frac{7 + 22}{2} = 14.5^\circ\text{C} = 287.5 \text{ K}$

from table A-2 (b) $C_{v,\text{avg}} = 0.716 + 0.002 \times \frac{287.5 - 250}{300 - 250}$

$$= 0.7175 \text{ kJ/kg K. OR 4}$$

$$\Rightarrow \Delta U = m C_v \Delta T = 113.45 \times 0.7175 \times (22 - 7)$$

$$\Rightarrow \Delta U = 1221.0 \text{ kJ}$$

$$\Rightarrow \Delta t = 1221.0 / 2.7 = 452.22 \text{ s} = 7.54 \text{ min}$$

(b) If truly constant volume then this would imply the room is air tight and pressure would increase as temperature increases... not true.

But not const. pressure closed system either (walls would have to move to maintain const. pressure).

In reality some of the air escapes during heating, so not truly 'closed'.

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But it would be a reasonable approximation since the temperature change is relatively small (and hence air loss, relatively small).

Question 4

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Question 4 – (9 marks)

Fill in the missing information in the following table for *refrigerant R134a*

	T (°C)	P (kPa)	h (kJ/kg)	x	Phase description
a.	36	912.35	250	0.88	saturated mixture
b.	50	500	291.98	—	superheated vapour
c.	-16	157.38	114.762	0.4	saturated mixture
d.	8.91	400	255.61	1.0	saturated vapour
e.	-32	76.71	10.09	0.0	saturated liquid
f.	-2	900	49.15	—	compressed liquid

1/2 mark
each
= 9 marks

$$(a) \quad h_f < h < h_g \Rightarrow \text{sat. mixture}, \quad x = \frac{h - h_f}{h_{fg}}, \quad P = P_{\text{sat}} = 912.35 \text{ kPa}$$

$$\Rightarrow x = \frac{250 - 102.34}{167.19} = 0.883$$

$$(b) \quad h > h_g \Rightarrow \text{superheated}, \quad \text{use Table A-13}, \quad P = 0.5 \text{ MPa}$$

$$h = 291.98 \text{ kJ/kg} @ \underline{50^\circ\text{C}}$$

$$(c) \quad x = 0.4 \Rightarrow \text{sat. mixture}, \quad P = P_{\text{sat}} = 157.38 \text{ kPa}$$

$$h = h_f + x h_{fg} = 30.67 + 0.4 \times 210.23 = 114.762 \frac{\text{kJ}}{\text{kg}}$$

$$(d) \quad \text{saturated vapour} \Rightarrow x = 1 \quad h = h_g @ 400 \text{ kPa} = 255.61 \frac{\text{kJ}}{\text{kg}}$$

$$T = T_{\text{sat}} = 8.91^\circ\text{C}$$

$$(e) \quad x = 0 \Rightarrow \text{saturated liquid at } T = -32^\circ\text{C}$$

$$\Rightarrow h = h_f @ -32^\circ\text{C} = 10.09 \text{ kJ/kg} \quad P = P_{\text{sat}} = 76.71 \text{ kPa}$$

$$(f) \quad h < h_f \Rightarrow \text{compressed (subcooled) liquid}$$

$$\Rightarrow h = h_f @ \underline{\text{temperature}} \rightarrow \text{look in table A-11}$$

$$h_f = 49.15 \frac{\text{kJ}}{\text{kg}} \text{ at } T = \underline{\underline{-2^\circ}}$$